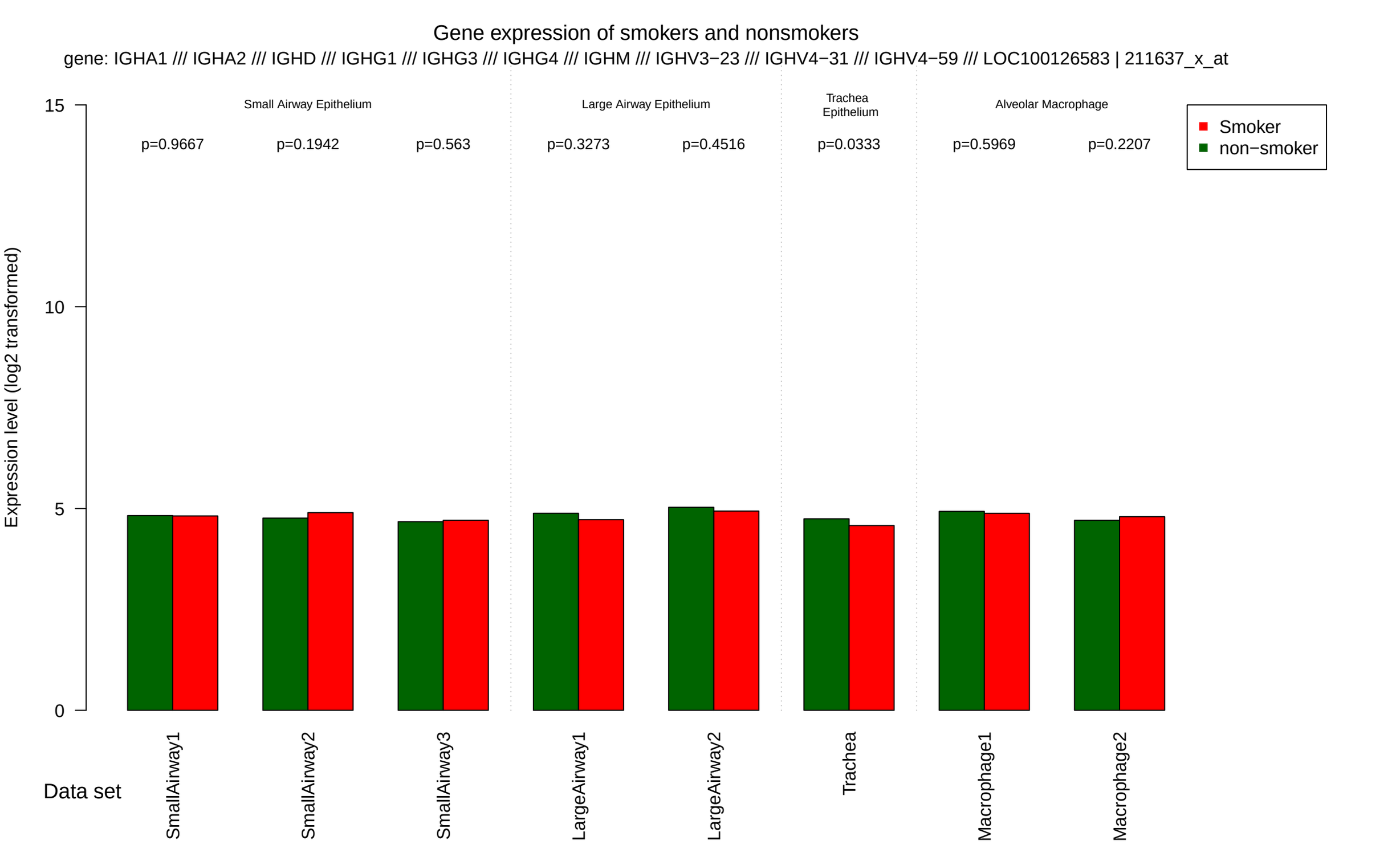
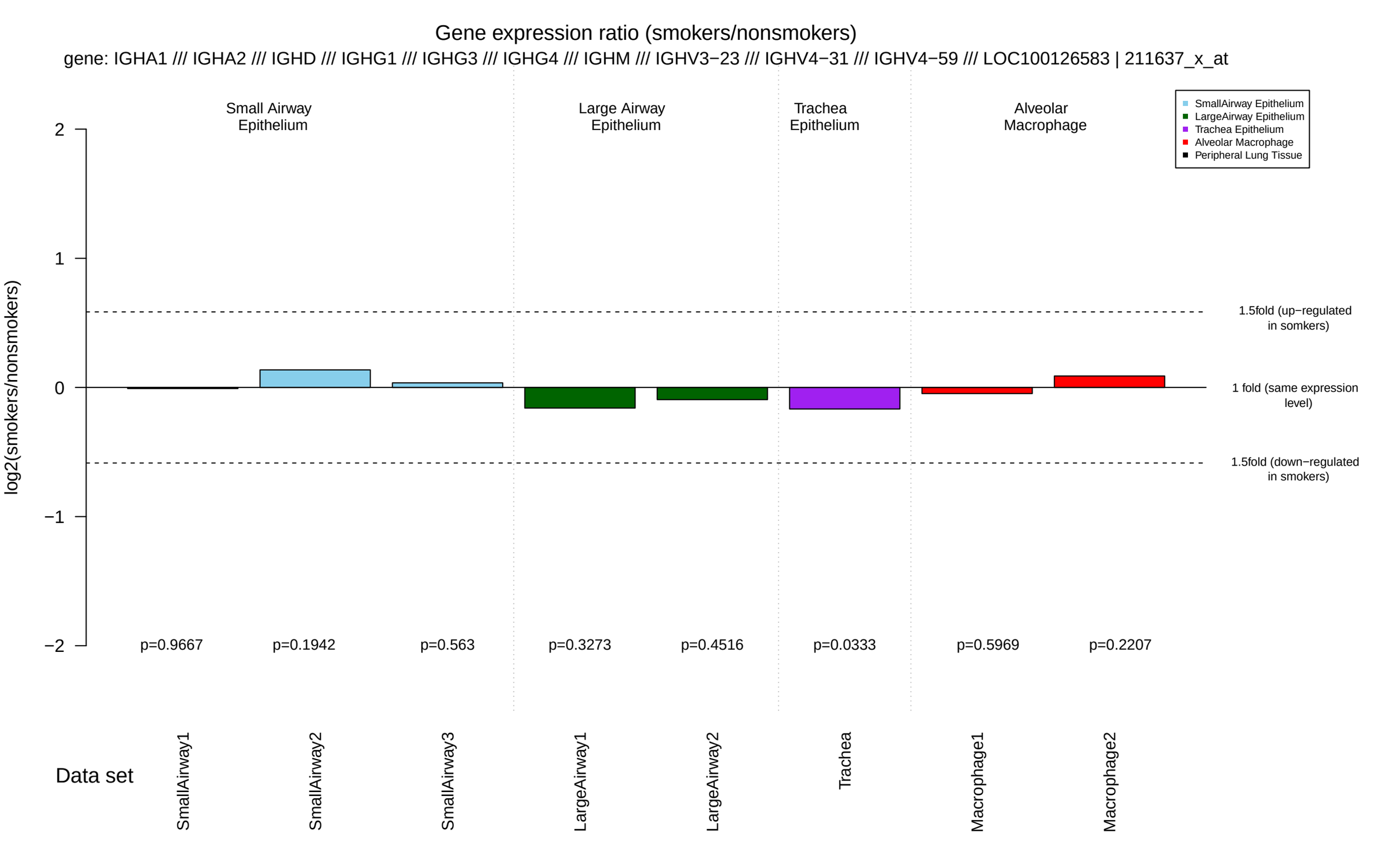
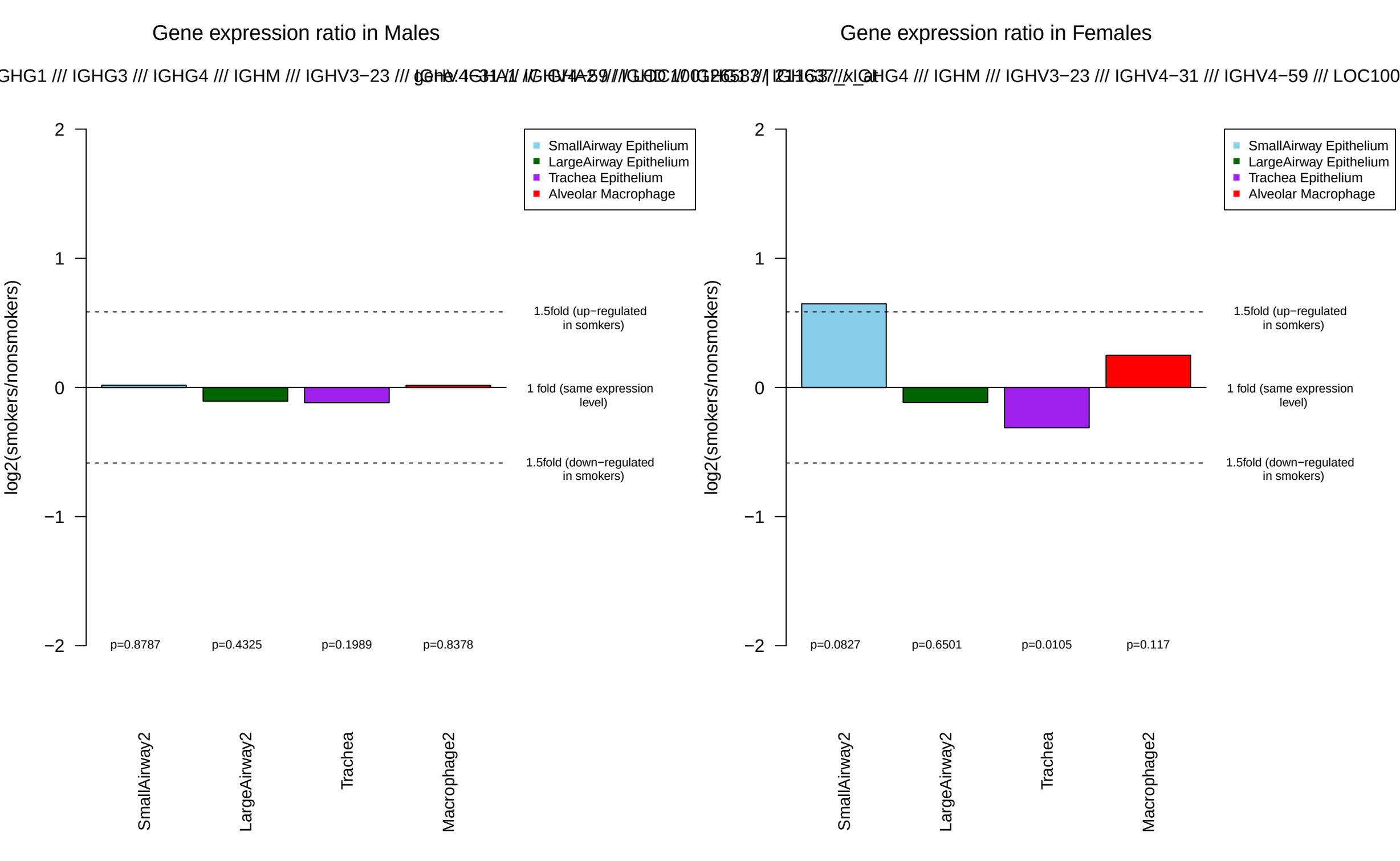
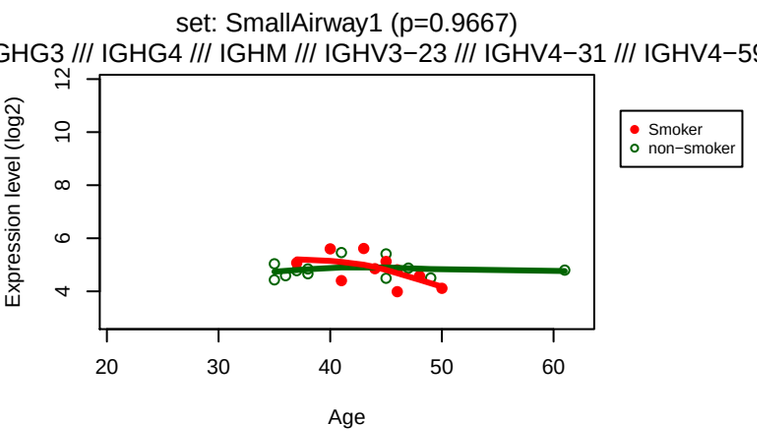
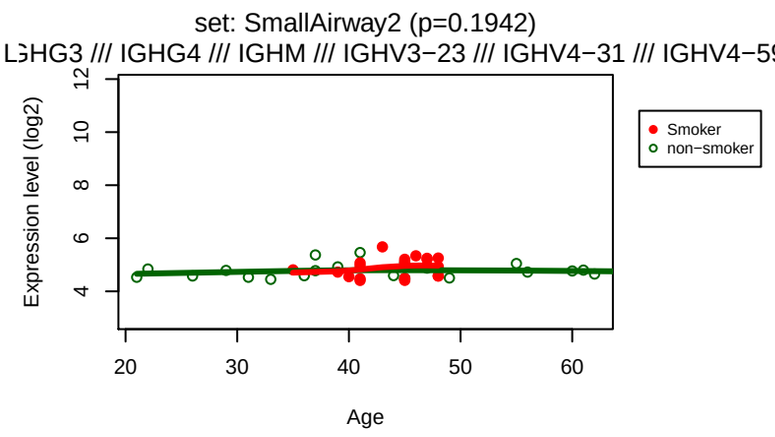
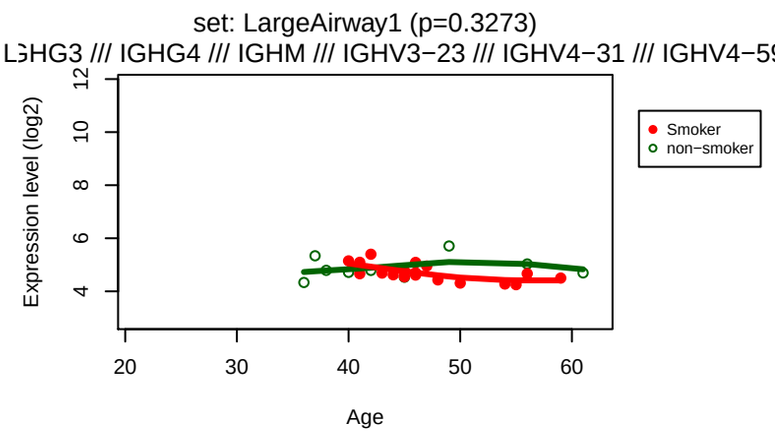

Gene expression of smokers and nonsmokers
gene: IGHA1 /// IGHA2 /// IGHD /// IGHG1 /// IGHG3 /// IGHG4 /// IGHM /// IGHV3−23 /// IGHV4−31 /// IGHV4−59 /// LOC100126583 | 211637_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.9667
p=0.1942
p=0.563
p=0.3273
p=0.4516
p=0.0333
p=0.5969
p=0.2207
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

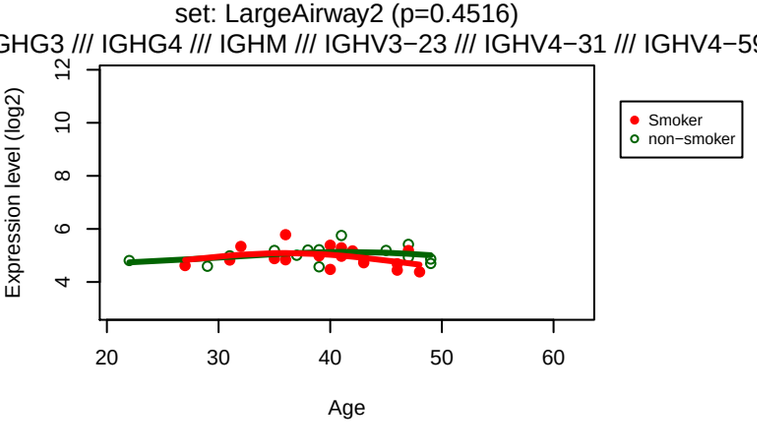

Gene expression ratio (smokers/nonsmokers)
gene: IGHA1 /// IGHA2 /// IGHD /// IGHG1 /// IGHG3 /// IGHG4 /// IGHM /// IGHV3−23 /// IGHV4−31 /// IGHV4−59 /// LOC100126583 | 211637_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9667
p=0.1942
p=0.563
p=0.3273
p=0.4516
p=0.0333
p=0.5969
p=0.2207
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

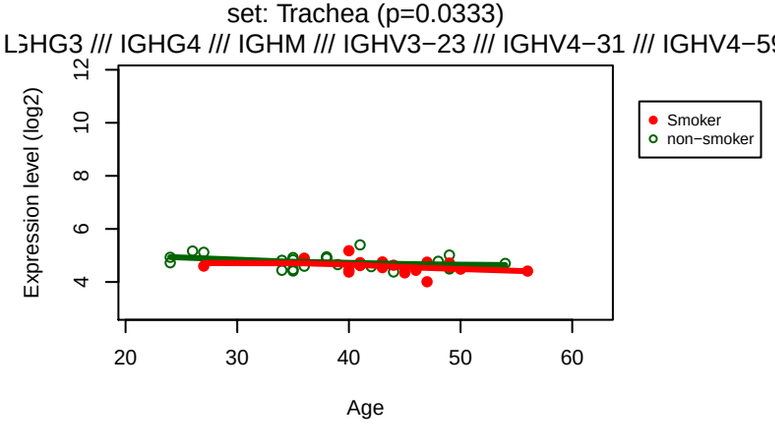

Gene expression ratio in Males
IGHG1 /// IGHG3 /// IGHG4 /// IGHM /// IGHV3−23 /// IGHV4−31 /// IGHV4−59 /// LOC100126583 /// 211637_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8787
p=0.4325
p=0.1989
p=0.8378
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
IGHG1 /// IGHG3 /// IGHG4 /// IGHM /// IGHV3−23 /// IGHV4−31 /// IGHV4−59 /// LOC100126583 /// 211637_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0827
p=0.6501
p=0.0105
p=0.117
SmallAirway2
LargeAirway2
Trachea
Macrophage2

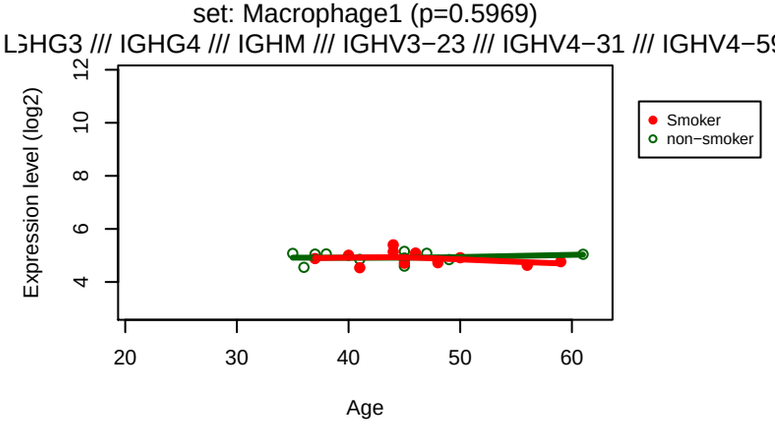

set: SmallAirway1 (p=0.9667)
GHG3 /// IGHG4 /// IGHM /// IGHV3−23 /// IGHV4−31 /// IGHV4−59 /// L
Expression level (log2)
Age
Smoker
non−smoker

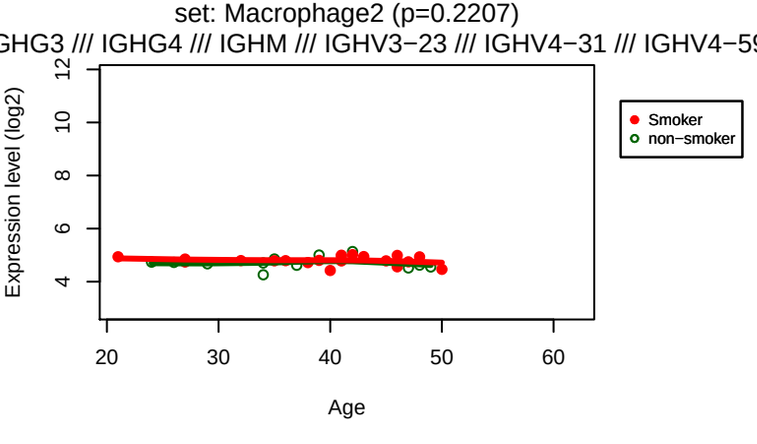

set: SmallAirway2 (p=0.1942)
GHG3 /// IGHG4 /// IGHM /// IGHV3−23 /// IGHV4−31 /// IGHV4−59 /// L
Expression level (log2)
Age
Smoker
non−smoker

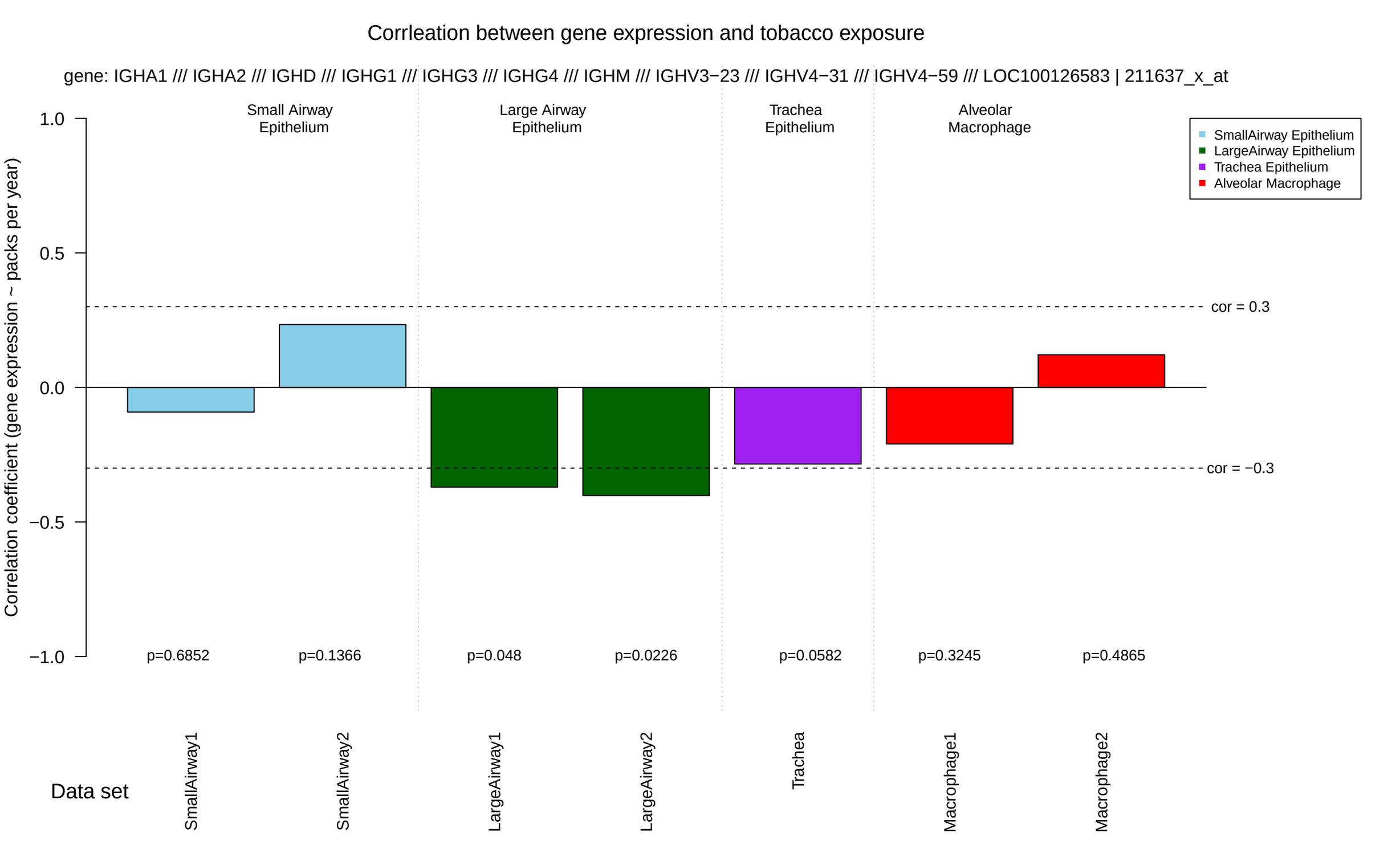

set: LargeAirway1 (p=0.3273)
GHG3 /// IGHG4 /// IGHM /// IGHV3−23 /// IGHV4−31 /// IGHV4−59 /// L
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4516)
GHG3 /// IGHG4 /// IGHM /// IGHV3−23 /// IGHV4−31 /// IGHV4−59 /// L
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0333)
GHG3 /// IGHG4 /// IGHM /// IGHV3−23 /// IGHV4−31 /// IGHV4−59 /// L
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5969)
GHG3 /// IGHG4 /// IGHM /// IGHV3−23 /// IGHV4−31 /// IGHV4−59 /// L
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2207)
GHG3 /// IGHG4 /// IGHM /// IGHV3−23 /// IGHV4−31 /// IGHV4−59 /// L
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IGHA1 /// IGHA2 /// IGHD /// IGHG1 /// IGHG3 /// IGHG4 /// IGHM /// IGHV3−23 /// IGHV4−31 /// IGHV4−59 /// LOC100126583 | 211637_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6852
p=0.1366
p=0.048
p=0.0226
p=0.0582
p=0.3245
p=0.4865
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2